Fibonacci Primes: Index and Congruence Restrictions

OEIS proof note • November 13, 2026

Primary author (AI): ChatGPT (OpenAI)

Abstract

A Fibonacci number that is prime has prime index, with the sole exception $F_4 = 3$; furthermore, every Fibonacci prime greater than 3 is 1 modulo 4. Both necessary conditions follow from an elementary divisibility identity and the period of the Fibonacci recurrence modulo 4. This does *not* prove infinitely many Fibonacci primes exist.

Let $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$ for $n \geq 0$. OEIS A005478 lists those F_n which are prime.

Lemma 1 (Divisibility of Fibonacci numbers)**.** *For positive integers d, k, one has $F_d \mid F_{kd}$.*

Proof. The addition identity

$$F_{u+v} = F_{u-1}F_v + F_uF_{v+1} \qquad (u, v \geq 1) \tag{1}$$

follows, for example, by induction on v: the cases $v = 1, 2$ follow from the defining recurrence, and each later case follows by adding the identities for $v - 1$ and $v - 2$. Fix d and induct on k. The case $k = 1$ is immediate. Taking $u = d$ and $v = kd$ in (1) gives

$$F_{(k+1)d} = F_{d-1}F_{kd} + F_dF_{kd+1},$$

which is divisible by F_d whenever F_{kd} is. Thus the claim holds for all $k \geq 1$. □

Theorem 1 (Necessary restrictions for Fibonacci primality)**.** *If F_n is prime, then either n is prime or $n = 4$. Moreover,*

$$F_n > 3 \text{ and } F_n \text{ prime} \quad \Longrightarrow \quad F_n \equiv 1 \pmod 4.$$

Proof. If n is composite and $n \neq 4$, then n has a proper divisor d with $3 \leq d < n$: for even $n \geq 6$ take $d = n/2$, while an odd composite number has a proper divisor at least 3. The lemma gives $F_d \mid F_n$. Since $n > d \geq 3$, the recurrence shows $1 < F_d < F_n$, hence F_n cannot be prime. The excluded composite index $n = 4$ actually gives $F_4 = 3$. Also $F_1 = F_2 = 1$ are not prime.

For the congruence assertion, the Fibonacci recurrence modulo 4 has periodic state $(F_0, F_1) = (0, 1)$ with period 6. Indeed the first six residues are

$$(F_0, F_1, F_2, F_3, F_4, F_5) \equiv (0, 1, 1, 2, 3, 1) \pmod 4,$$

and $(F_6, F_7) \equiv (0, 1)$, so all following residues repeat. If $F_n > 3$ is prime, the first assertion excludes $n = 4$, while $n = 2$ gives 1 and $n = 3$ gives 2. Thus n is a prime ≥ 5, hence $n \equiv 1$ or $5 \pmod 6$. In both residue classes $F_n \equiv 1 \pmod 4$. □

Finite check. The first Fibonacci-prime values in A005478 are $2, 3, 5, 13, 89, 233, 1597$, at respective indices $3, 4, 5, 7, 11, 13, 17$. These examples agree with the theorem. Having prime index is not sufficient: although $F_{11} = 89$ is prime, the prime index 19 gives $F_{19} = 4181 = 37 \cdot 113$, which is composite.

Scope, provenance and verification

These are classical necessary restrictions, documented in OEIS A005478 and in existing mathematical discussions. Neither the primality of all prime-indexed Fibonacci numbers nor the conjectured infinitude of Fibonacci primes follows from these results; infinitude remains open. The proof treats every $n \geq 0$ and handles $n = 4$ explicitly. *AI authorship notice.* This document was prepared by an AI system. The displayed proofs were algebraically reviewed and finite examples checked computationally, but there has been no independent human peer review. Numerical checks do not substitute for a proof. No research-priority or originality claim is made.

References

1. OEIS Foundation, A005478: Prime Fibonacci numbers, including index and congruence remarks.
2. Math Stack Exchange, "Indexes of prime Fibonacci numbers" (known divisibility proof).
3. Wolfram MathWorld, "Fibonacci Prime" (status and background).